



Passive infrared thermography technique for concrete structures health investigation: case studies

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Received: 15 November 2022 / Accepted: 4 January 2023
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Abstract

Non-destructive testing techniques are often employed for periodic maintenance and reparation of construction works, especially for large-scale structures. These techniques can be applied during construction or even on existing structures. Out of the non-destructive testing techniques, infrared thermography is a powerful technique to investigate structural conditions and locate the damages because of the ability to visualize unseen subsurface delamination through thermal images. The technology of infrared detectors has significantly developed with high-quality products and acceptable prices. Therefore, the applications of the technique are feasible. This article introduces current infrared thermography techniques. Case studies have been implemented using passive infrared thermography for investigating construction works, such as buildings, warehouses, and bridges. The study has elucidated the applications of the method in detecting moisture penetration, area of insulation, energy loss, and subsurface delamination. A guide to passive thermography has been recommended for site inspection. Finally, common errors during thermal surveying are discussed to help engineers reduce unnecessary mistakes.

Keywords Passive infrared thermography · Non-destructive testing · Concrete structures · Emissivity

Introduction

Structural health investigation (SHI) or structural health monitoring (SHM) is the overall target, using destructive and non-destructive testing (NDT) for initial evaluation. Condition monitoring (CM) using sensors is then applied for ongoing assessment (Kaveh & Eslamlou, 2019; Kaveh et al., 2022). Thus, preliminary testing is essential to locate potential deteriorations, especially for unseen defects. The infrared thermography (IRT) technique for structural health monitoring appeared just around the 1950s, although the infrared (IR) ray was discovered earlier in 1800 by astronomer Sir Frederick William Herschel (Maldague, 2001; Prakash Rao, 2008). The terms IRT, thermographic testing, thermal imaging, or simply thermography indicate the NDT imaging technique (Castanedo & Tarpani, 2013), which is

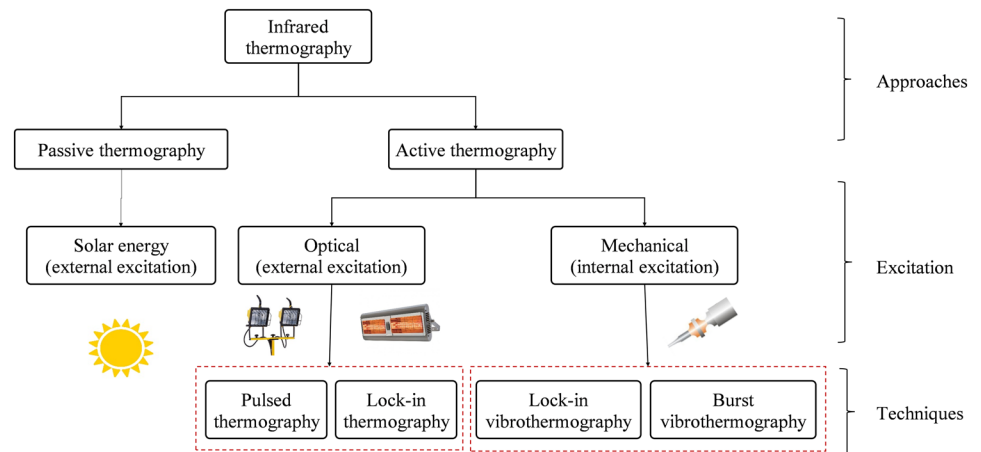
formally defined as “a non-destructive, non-intrusive, non-contact technique” that allows mapping of thermal patterns (also known as thermograms) on the surface of objects, bodies or systems through using an IR imaging instrument (so-called IR thermal camera or IR camera/detector/imager) (Maldague, 2001).

There are two common approaches to the IRT techniques: active thermography (AIRT) and passive thermography (PIRT), as shown in Fig. 1. Active thermography (or dynamic) is a process that an external source heats the structure to obtain temperature contrast. There are three techniques of optical excitation mode: pulsed thermography (local heating, using flash lamps or heat gun, denoted as PT), lock-in thermography (heating the object surface periodically by halogen lamps, denoted as LT), and pulse phase thermography (repeated heating at short intervals of time, this technique combines features of PT and LT); and vibro-thermography technique, which uses ultrasonic waves (amplitude modulated or pulse) to excite internal features (Castanedo et al. 2007; Maldague et al., 2002; Milovanović & Banjad Pečur, 2011; Tran et al., 2018a, 2018b; Wićek & Poksinska, 2006). Table 1 shows a comparison of the active thermography techniques. Otherwise,

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Fig. 1 Infrared thermography techniques**Table 1** Summarized active thermography techniques (Castanedo et al., 2007; Milovanović & Banjad Pečur, 2011; Więcek & Poksinska, 2006)

Technique	Passive infrared thermography	Active infrared thermography (AIRT)		
		Pulsed thermography	Lock-in thermography	Vibro-thermography
Denoted as	PIRT	PT	LT	VT
Heat source	Solar radiation	Heat pulse	Periodic thermal waves	Mechanical vibrations to thermal energy
Regime		Pulsed	Modulated	Pulsed/modulated
Equipment	None	-IR camera -Flash lamps -PC and control unit	-IR camera -Halogen lamps -PC and control unit	-IR camera -Ultrasonic transducer -PC + control unit
Advantage	-Natural heat source	-Fast and easy to deploy -Shallow inspection	-Depth inspection -Low-power thermal waves	-Detecting flaws hardly visible -Inspecting large structural areas -Faster than PT
Disadvantage	-Depending on the weather conditions, ex. cloudy, rainy, wind speed, etc	It is affected by non-uniform heating	It requires a test for every inspected depth. Hence, it is slower than PT	Mechanical loading is sometimes hard to achieve

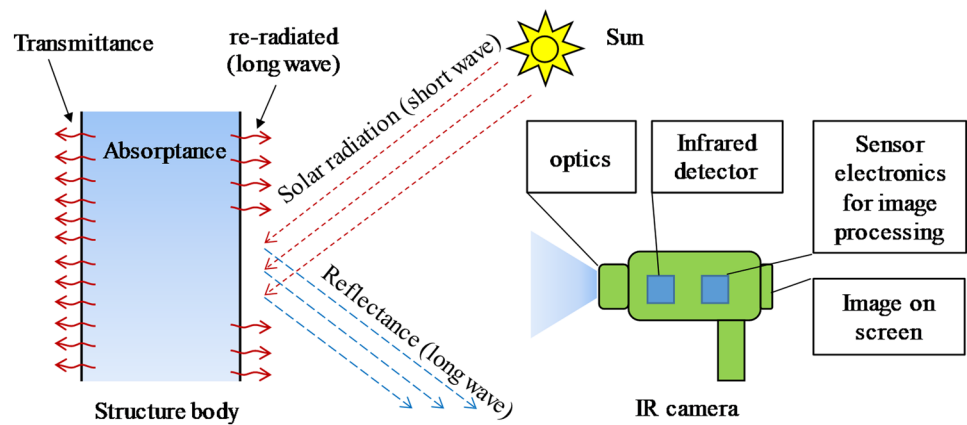
passive thermography (or static), in which the structure is heated by a natural source (solar radiation). It operates in two wavelength ranges: mid wavelength (3–5 μm , denoted as MWIR) is generally used to visualize temperature between 40 and 2000 $^{\circ}\text{C}$, and long wavelength (8–14 μm , denoted as LWIR) is applied for temperature from -20 $^{\circ}\text{C}$ to the ambient temperatures. To use IRT techniques effectively, the following parameters should be considered: surface properties, material, air temperature, humidity, and live organisms (i.e., bacteria, mushrooms) (Maldague et al., 2002; Więcek & Poksinska, 2006).

Generally, AIRT techniques are suitable for detecting small areas. The AIRT experiment is usually done in a laboratory to avoid the influence of the wind; thus, it provides a higher quality thermal image than PIRT techniques. However, AIRT techniques require equipment and well-trained technicians for operation, and they have a high cost. The PIRT technique has been proven effective in investigating

large-scale structures because of its low cost and time consumption. This study applies the PIRT to locate the subsurface defects. Practical case studies for buildings, a warehouse, and a bridge, are analyzed and discussed.

Thermal physics and infrared camera works

Solar radiation is incident on the body of a structure, some of it is transmitted, absorbed, and some is reflected, as illustrated in Fig. 2. The emitted energy from the body depends upon the surface temperature and surface characteristics (Holst, 2000). For an IR detector, the infrared energy (reflected radiation and absorbed radiation) coming from the object is focused by the IR detector's optics. The detector sends the information to sensor electronics for image processing. The electronics translate the data from the detector into an image that can be viewed on the screen (FLIR, 2011).

Fig. 2 Principle of an IR camera as obtaining solar radiation

An object reacts to incident radiation from its surroundings by absorbing, reflecting a portion of it, or allowing some of it to pass through (i.e., as through a camera lens). The total radiation law (W in W/m^2) is derived from Eq. (1) (FLIR, 2012).

$$W = \alpha \cdot W + \rho \cdot W + \tau \cdot W. \quad (1)$$

The relationship between the three coefficients can be simplified to: (Holst, 2000):

$$\tau + \alpha + \rho = 1, \quad (2)$$

where τ , α , and ρ are the transmittance, absorptance, and reflectance, respectively. The thin plastics are transmissive, but in most cases, e.g., metals, woods, and concrete, the transmittance is zero, then Eq. (1) can be expressed for concrete structures as follows:

$$\alpha + \rho = 1. \quad (3)$$

The equation for radiation of an object according to the Stefan–Boltzmann equation is:

$$W = \varepsilon \cdot \sigma \cdot T^4, \quad (4)$$

where ε is the emissivity, σ is the Stefan–Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$), and T is the temperature (measured in Kelvin). The temperature to the fourth power in Eq. (4) shows that a small temperature change produces a significant change in energy emitted.

When an object is in thermal equilibrium with its environment, the energy absorbed must equal the amount radiated. Thus, the absorptance (α) equals the emissivity (ε) (Holst, 2000). Hence, Eq. (3) is $\varepsilon + \rho = 1$. The emissivity (ε) is defined as a rate of $W_{\text{actual}}/W_{\text{blackbody}}$ (Planck's law), depends upon the material, its surface condition (may be polished, oxidized, or coated; or the age of the paint, dust, dirt, and condensation), and temperature (Holst, 2000). Table 2 provides the emissivity for normal incidence applied in IRT investigation. It is noted that if the wrong

Table 2 Emissivity for materials usually used for civil engineering structures (American Concrete Institute, 2013; FLIR, 2012; Holst, 2000; Vollmer & Mollmann, 2010)

Material	Material	Temp. (°C) ^a	Wavelength ^b (μm)	Emissivity
Concrete	Concrete	20		0.92
	Dry concrete	36	5	0.95
	Rough concrete	17	2–5.6	0.92–0.97
	Mortar	17	2–5.6	0.87
	Dry mortar	38–260	2–5.6	0.94
Granite	Natural surface	36	5	0.96
	Smooth	21		0.45
Brick	Common	17	2–5.6	0.81–0.86
	Firebrick	17	2–5.6	0.68
	Fireclay	1000		0.75
	Masonry	0	5	0.94
Wood	Plywood	17	2–5.6	0.83–0.98
Steel	Cold rolled	93	5	0.75–0.85
	Galvanized			0.28
	Polished sheet	38	5	0.07
	Rough surface		2–5.6	0.96

^aIf the temperature is not given, assume that the values are for ambient conditions (77°F or 25 °C)

^bIf the wavelength is not specified, the emissivity values can be used for both MWIR and LWIR

emissivity is entered in the IR camera, it will lead to false temperature results.

For selecting an IR camera, a shortwave camera (3–5 μm) in typical measurement environments generally offers better temperature measurement accuracy than a longwave camera. Meanwhile, long-wavelength cameras are reasonably priced and meet the accuracy requirements of civil engineering (Clark et al., 2003). For inspection purposes of building and bridge structures, a long-wavelength (from 8 to 14 μm waveband) camera is recommended.

Case studies in structural health monitoring

Case study no. 1: practical investigation for building structures

Two concepts of thermal inspection are “warning” and “cooling” scan (IAEA, 2002). The warning condition (material is absorbing the external heat) is the survey during day time, whereas the cooling (material is re-radiating) condition is from evening until early morning. However, “cooling” or “warning” might depend on the weather conditions (i.e., ambient temperature, humidity, wind speed, etc.). For this practical survey, the test is “cooling” condition. In this study, three buildings and a warehouse at Chonnam National University in Yeosu, South Korea, have been investigated indoors and outdoors. The inspection includes detecting moisture penetration, insulated areas, and energy loss. The test was carried out at around 3:00 p.m., that morning was a sunny day, but at the time of testing, it was mostly cloudy with the air temperature of 21.7 °C, wind direction on the south-southeast at 11.3 km/h, and humidity of 42%. These buildings are made of reinforced concrete. Thus, the emissivity (ϵ) is usually

about 0.95, suitable for dry concrete, rough granite, and mortar materials, as recommended in Table 2.

Finding moisture penetration

Moisture can lead to severe problems for structures and health issues. It exists in roofs, walls, or floors in internal and external structures for many different causes. The thermal survey can easily and quickly discover where water leakage sources or water infiltration in flat roofs after rain. The presence of moisture causes lower temperatures because of ambient evaporation (Prakash Rao, 2008). As shown in Fig. 3, taken at building 1, the bright color in the thermal image has a temperature range of 22.1–22.5 °C, except the dark color shows lower temperatures (approximately 20.8 °C) because of the penetrating of moisture. There was no standing water above the floor. The water may have infiltrated from the adjacent drain hose due to a damaged waterproofing agent.

In another part of building 2, it looks like water is standing on the concrete slab, as seen in Fig. 4a. There was heavy rain 4 days ago; hence, it is easy to predict that the waterproof ability of the upper surface of this ceiling was not working well. However, when detecting by IR thermal camera, the temperature pattern has no distinction to adjacent

Fig. 3 Moisture detecting for the concrete slab (building 1)

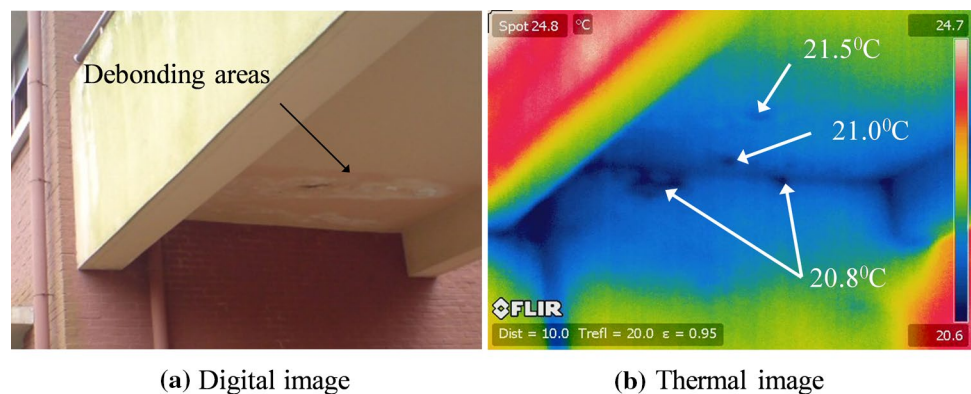
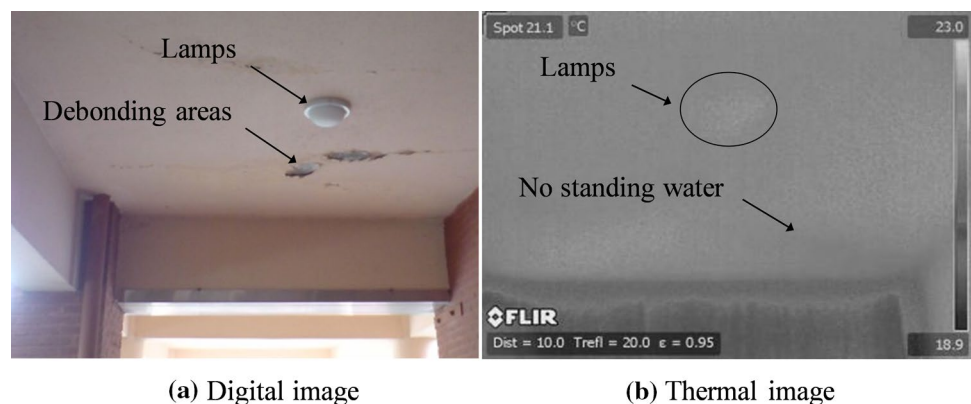


Fig. 4 No water standing in the concrete slab (building 2)



areas (the average concrete surface temperature is about 21.4 °C). Therefore, this section of the concrete slab has no moisture penetration. If the thermal survey is not applied, it might lead to the wrong method to fix this problem; thus, it will cost more budget to repair these areas.

Detection of insulated areas and energy loss

The thermal camera can detect mold and poorly insulated areas. It is effective with a sandwich construction (e.g., concrete–insulation–concrete wall), the wall and roof of warehouses. Figure 5 indicates that the ceiling panels (bright color) of building 3 have a higher temperature than the surrounding walls, with a temperature difference of 2 °C approximately. Hence, we can believe that the reinforced concrete slab above the ceiling panels has not fully insulated the outside solar radiation. It is noted that the above roof is made of reinforced concrete. For cities with extreme weather conditions, the energy loss is significant because we have to use much energy for cooling the indoor atmosphere on a hot day or much power for heating to keep it warmer in freezing weather. With this viewpoint, IRT technology plays a vital role in overcoming the energy efficiency problem for buildings.

Fig. 5 Insulation survey for building ceiling (building 3)

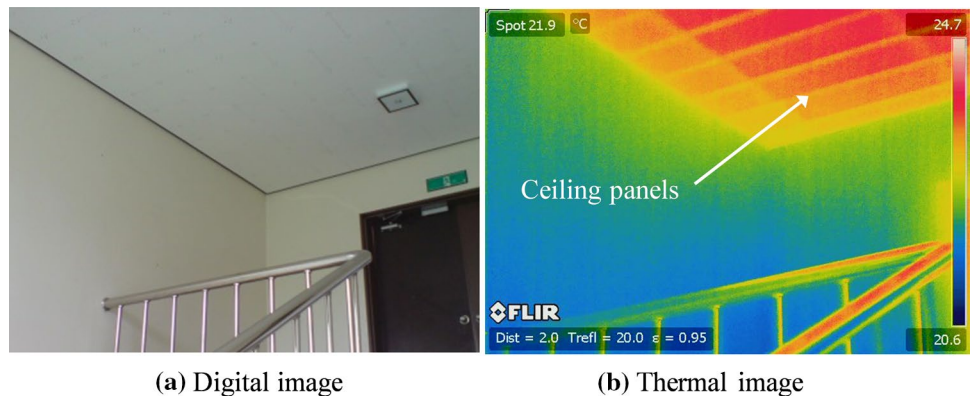
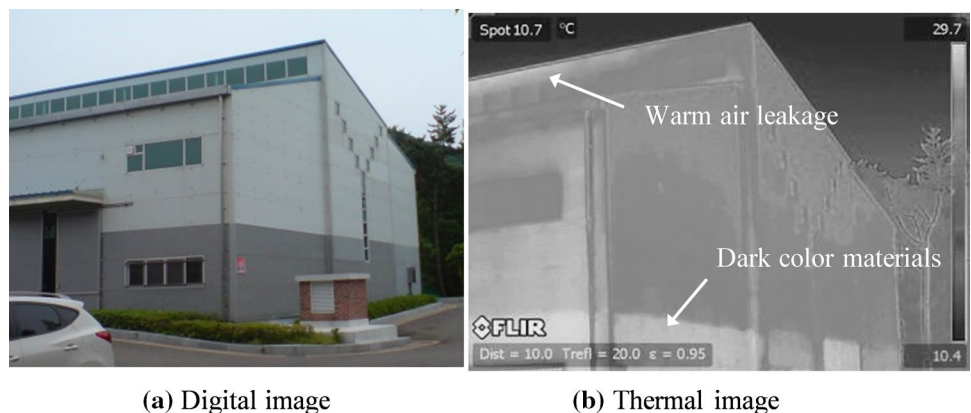


Fig. 6 Detecting the energy loss for a warehouse



For inspection of a steel structure warehouse (Fig. 6), the temperature pattern is relatively homogeneous except for the sections having dark color material because they have different emissivity coefficients. There might be energy loss from the joints between the wall and the roof (a bright color strip) caused by warm air escaping from inside to outside. Notably, at the bright color strip in Fig. 6, the temperature surface is around 28 °C, whereas the surrounding areas are just from 25 to 26 °C. In general, the IR thermal camera can detect the quality of insulation materials in old buildings or check for acceptance of insulation materials after installation in a new building.

Case study no. 2: practical investigation for bridge structures to locate delaminations

Figure 7 shows the application of PIRT for an existing bridge. Haechang bridge, with a total length of 192 m, an overall width of 5 m, and a gravity-type pier height of 3–4 m approximately, was built by reinforced concrete in 1984 in South Korea. There are 16 spans with a maximum span of 12 m. In this inspection, the piers and concrete deck of the bridge were investigated. Some questionable places exposed rebars at pier P9, as shown in Fig. 7b. However, it is impossible to know by the naked eye the degree of delaminations

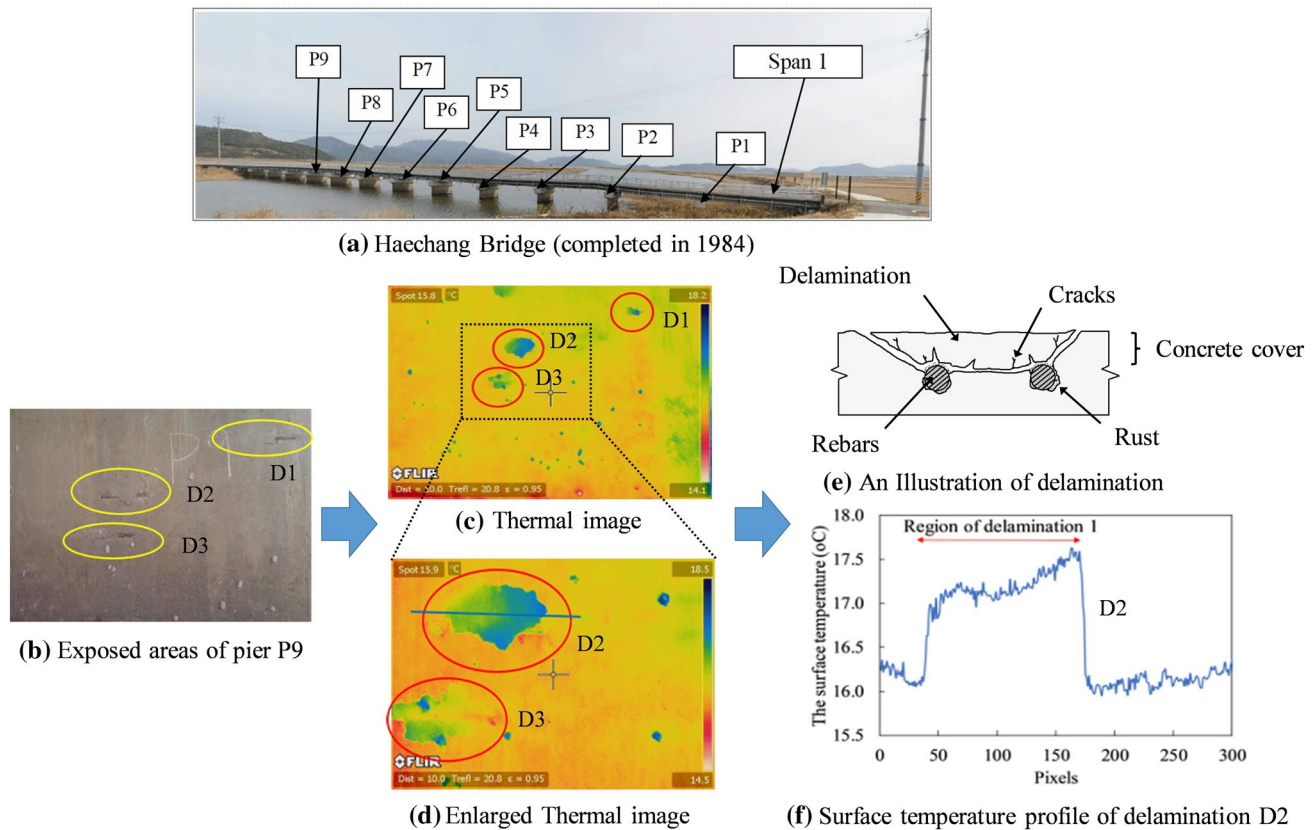


Fig. 7 Detecting delaminations for an existing bridge

in the surrounding areas. Delamination is defined as a laminar discontinuity, such as an area of unbonded materials (ASTM D4580-03, 2007; Pailes & Gucunski, 2015). Under PIRT scanning results, the delaminated areas are clearly seen based on the difference in surface temperature between the defective and non-defective areas. In particular, the thermal image shows the delaminated areas at defects D2 and D3. A longitudinal profile crossing the large delamination D2 (Fig. 7f) shows that the surface temperature above the defective area is 17–17.5 °C. In comparison, the background temperature (non-defective area) is just 16.3 °C approximately.

These delaminations can be determined qualitatively by traditional methods such as chain dragging (for horizontal surfaces) or tapping by hammer (for both vertical and horizontal surfaces) (American Concrete Institute, 2013; Washer, 1998). Thanks to the advanced technology, the PIRT provides a quantitative evaluation rapidly and precisely to locate the delaminations. In some cases, scanning by an infrared thermal camera might not indicate any defects or evidence of potential problems because of many factors. For greater efficiency, thermography testing should be applied with other methods such as sonic tests, ground penetrating radar (GPR), or ultrasonic pulse velocity (UPV) (Pollock et al. 2008; Maser, 2009).

Discussion on the investigation procedure and factors affecting thermal imaging

Process of carrying out thermal inspection

The following four steps for thermal investigation should be considered based on our experiences testing the IRT projects (Mac et al., 2019; Tran, 2021), as demonstrated in Fig. 8.

Step 1—Data collection: collect the design and as-built documents of the surveyed construction works to discover the types and properties of materials and construction method statements; check the weather (e.g., ambient temperature, visibility, humidity) of the expected day and several previous days.

Step 2—Preliminary survey at the site

- Outdoor check: direction of sunlight, wind speed, obstacles such as trees, and adjacent high buildings, which might shade on or reflect the inspected structures. These issues influence the thermal pattern.
- Indoor check: bookshelves or pictures hanging on the wall might change the thermal pattern. Therefore, all items from the wall should be removed at least 6 h before inspection (FLIR, 2011). Besides, reflections (e.g., glass

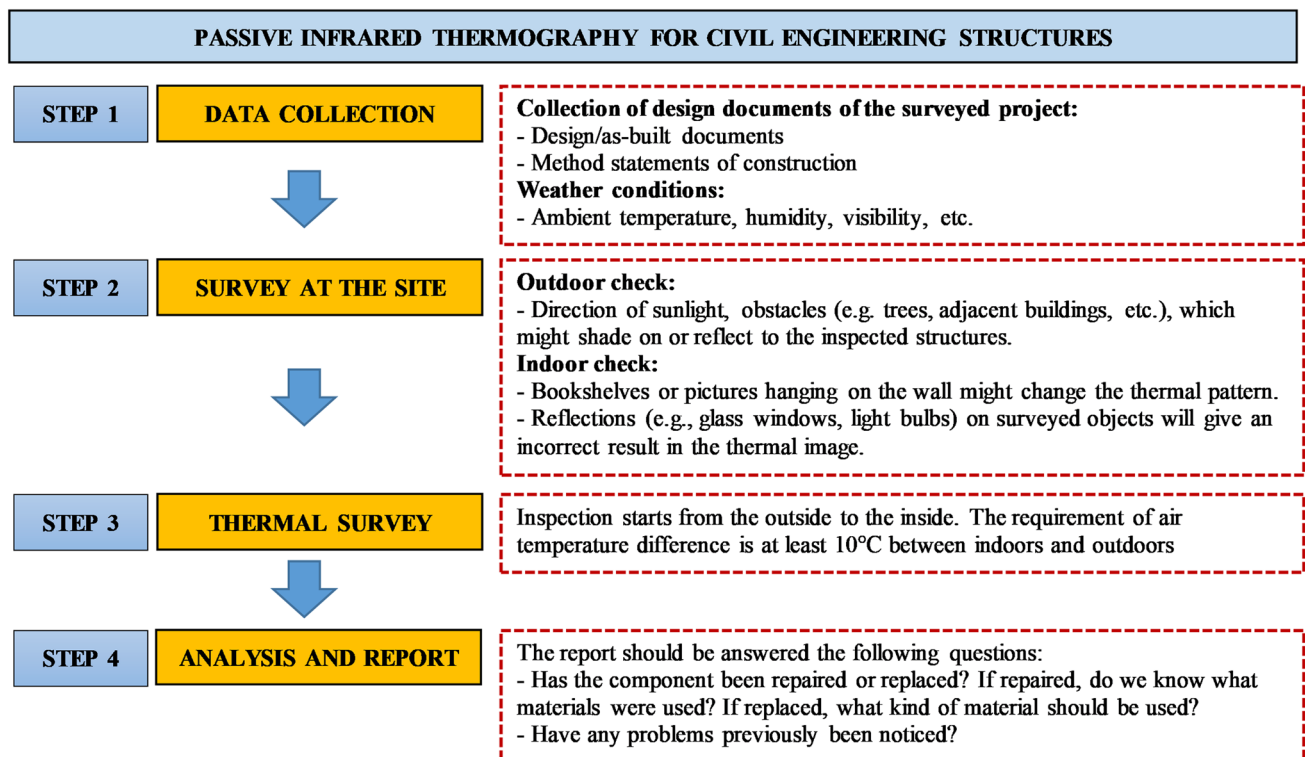


Fig. 8 Proposal process of passive infrared thermography inspection

windows, light bulbs) on surveyed objects will give an incorrect result in the thermal image.

Step 3—Thermal survey: inspection starts from the outside to the inside. The requirement of air temperature difference is at least 10 °C between the indoors and outdoors (FLIR, 2011).

- Outdoor inspection: same as step 1 but with more detail.
- Indoor inspection: moving furniture out and removing drapes 6 h before the inspection. In addition, checking the heating systems and air conditioning systems that can cause misleading thermal patterns (FLIR, 2011).

Step 4—Analysis and reporting: the report should answer the following questions: Has the component been repaired or replaced? If repaired, do we know what materials were used? If replaced, what kind of material should be used? Have any problems previously been noticed? (Holst, 2000).

Factors affecting thermal imaging

The measured temperatures are influenced by factors: subsurface configurations, surface conditions, environment, and humans. The heat flow can only be blocked or decreased by the insulating effects of the anomalies of

testing material (e.g., air void, delamination). Hence, air void has a lower thermal conductivity than the surrounding concrete (IAEA, 2002). Besides, when the re-radiated waves are emitted from the object's body, they might be stopped by defects. Therefore, the outers surface with existing defects have a lower temperature than other areas, as illustrated in Fig. 3.

The second factor is surface conditions. The emissivity value is higher for rough surfaces and lower for smooth surfaces. For instance, rough concrete has an emissivity range of 0.92–0.97, while polished steel sheet has an emissivity of about 0.07, as seen in Table 2. Practically, the engineer must be aware of different surface textures, e.g., textured surfaces, oil spots, or dirt on the surface (Prakash Rao, 2008; Więcek & Poksinska, 2006).

The environment is the third factor, such as clouds, wind, surface moisture, and nearby obstacles, as seen in Table 3 for detail. Humans, such as distance observations and camera viewing angles, might cause the last factor.

- Viewing angle of the camera: to avoid reflection of the camera, a recommended viewing angle is about 5°–60° (where 0° is perpendicular) (FLIR, 2011).
- Distances observation depends on the lens's capacity (e.g., a telephoto lens for a long shoot). An excellent thermal image should have a resolution of 640 × 480 pixels

Table 3 Factors of the environment (FLIR, 2011; Holst, 2000; IAEA, 2002; Więcek & Poksinska, 2006)

Factors	Description/reasons	Recommended
Solar radiation	Change in solar irradiance over time	Solar irradiance is required 700 W/m ² (at least 500 W/m ²)
Cloud cover	Clouds will reflect infrared radiation, slowing the sky's heat transfer process	Night-time testing should be performed during times of little or no cloud cover to allow the most efficient energy transfer out of the concrete
Temperature difference	The low-temperature difference leads to a bad result in the thermal image	A temperature difference of at least 10 °C is advisable
Wind speed	A high gust of wind might reduce surface temperatures	For qualitative measurements, the image should be taken at wind speeds of less than 25 km/h
Surface moisture	Moisture tends to disperse the surface heat	The test should not be performed while the structure surface is covered with standing water
Partial shadowing (e.g., trees)	Lead to a partial area of the surface cooler than other areas	It should be considered while analyzing thermal images
Reflection	(from adjacent buildings, sun, clouds)	Should be considered while surveying and analyzing thermal images

(at least 320×240 pixels), referred to as a FLIR camera (FLIR SC660 Catalog, 2014).

Conclusion

The studies have explored the applications for many types of civil engineering structures using passive infrared thermography. Construction works including buildings, warehouses, and bridges, were detected for moisture penetration, insulation, energy loss, and the subsurface delamination between concrete cover and rebars. A guideline for using passive thermography has been suggested for indoor and outdoor inspections. Finally, some common mistakes during thermal surveying were discussed, including solar radiation, cloud cover, temperature difference, wind speed, surface moisture, partial shadowing, and reflection from an adjacent building, to help field engineers improve their testing results.

Acknowledgements Grants supported this research funded by Nha Trang University (No. TR2020-13-23), Korean government (MSIT) (No. 2021R1A2C1005587) and the contributions of the research team members in the SA-NDE-T Lab at the Civil Engineering Department of Nha Trang University.

Author contributions QHT and JH: conceptualization, methodology, formal analysis, software, data curation, investigation, writing—original draft, visualization, validation. QMD, XTP, TCT, and TXN: software, data curation, visualization. All authors reviewed the manuscript.

Funding The authors have not disclosed any funding.

Availability of data and material The data that supports the finding in this study are not publicly available. Any access request should be sent to huytq@ntu.edu.vn.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval and consent to participate The authors have no conflict of interest to declare relevant to this article's content.

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